Thinking Mode Induces Confidence Compression in Reasoning LLMs:
A Pre-Registered Type-2 SDT Analysis

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Pre-registered: OSF (osf.io/2jwub) | Code: github.com/synthiumjp/reasoning-metacog

Abstract

Reasoning LLMs generate explicit thinking traces before answering, improving accuracy on knowledge-intensive tasks. We tested whether thinking mode also improves the discriminative power of token-probability-derived confidence signals -- the ability of normalised log-probability (NLP) and first-token softmax to separate correct from incorrect responses. In a pre-registered study, three 7-8B reasoning models (Qwen3-8B, DeepSeek-R1-Distill-Qwen-7B, DeepSeek-R1-Distill-Llama-8B) answered 2,000 TriviaQA items in thinking and non-thinking mode, yielding 12,000 primary trials. Metacognitive discrimination was assessed via Type-2 AUROC ($AUROC_2$), M-ratio, and four binding diagnostics. The Qwen3-8B result is confirmatory (all diagnostics pass); R1-Distill results are descriptive (D3 monotonicity failure in thinking mode). Thinking mode improved accuracy (+10.6 pp for Qwen3-8B) but significantly reduced NLP-based $AUROC_2$ in all three models (Qwen3-8B: Delta-$AUROC_2$ = -0.231, 95% CI [-0.262, -0.198]). First-token softmax $AUROC_2$ also degraded in all three models, suggesting the effect is not specific to mean log-probability. The evidence is consistent with NLP compression as the primary mechanism: thinking-mode NLP variance dropped to 16% of non-thinking levels, collapsing Cohen's *d* for correct/incorrect separation from 1.02 to 0.14. A D1 decomposition revealed model-dependent pathways: in Qwen3-8B, the degradation was driven primarily by answer-distribution shift, partially offset by trace-conditioned context recovery; in R1-Distill models, the trace context itself actively harmed discrimination. Across models and diagnostics, thinking mode induces a confidence homogenisation regime: responses become more accurate on average but less separable in confidence space, reducing the utility of token-level signals for per-trial reliability assessment.

1 Introduction

Reasoning-capable LLMs generate explicit thinking traces -- extended chain-of-thought sequences enclosed in tags -- before producing a final answer. This process reliably improves accuracy on mathematical, logical, and knowledge-intensive tasks (DeepSeek-AI, 2025; Qwen Team, 2025). A natural question follows: does thinking mode also improve the model's ability to distinguish its correct from incorrect responses?

In signal detection theory, metacognitive discrimination is quantified by Type-2 sensitivity: how well a confidence signal separates correct from incorrect responses. In LLMs, normalised log-probability (NLP) of the answer tokens serves as an implicit confidence signal. Prior work in the Classical Minds, Modern Machines (CMM) programme established that NLP-based M-ratio (the ratio of Type-2 to Type-1 sensitivity; Maniscalco & Lau, 2012) is a valid decomposition of LLM confidence across 224,000 trials (Cacioli, 2026a; arXiv:2603.25112), though format-dependent (Cacioli, 2026c; arXiv:2604.08976). Non-parametric $AUROC_2$ provides a complementary, format-robust measure of discrimination (Cacioli, 2026c).

Two prior findings motivate the present study. First, Yoon et al. (2025) reported that thinking mode improves calibration (lower ECE) in reasoning LLMs. Calibration and discrimination are mathematically independent; lower ECE does not entail higher $AUROC_2$ (Cacioli, 2026a). Second, Cacioli (2026d; arXiv:2604.22215) found that verbal confidence in 3-9B models is categorically compressed to ceiling under minimal elicitation, with reasoning-trace length negatively correlated with verbalised confidence (rho = -.36) in a distilled reasoning model. Whether implicit confidence signals show analogous compression under thinking mode is unknown.

1.1 The construct validity question

We are careful to distinguish two claims. A strong claim is that thinking mode degrades *metacognition* -- the model's internal capacity to monitor its own accuracy. A weaker but more defensible claim is that thinking mode degrades the *discriminative utility of implicit confidence signals* -- NLP and first-token softmax become less informative about correctness. Our evidence supports the weaker claim. The stronger claim would require ruling out that the signals are degraded solely by the generative regime change rather than by a change in internal monitoring, which our design cannot fully adjudicate.

1.2 The context-conditioning threat

In thinking mode, the final answer is conditioned on the thinking trace. NLP of the answer therefore reflects both uncertainty and the local token predictability created by the preceding trace. This is the primary threat to validity. An abridged-NLP diagnostic (D1) addresses prefix-level effects by recomputing NLP in fresh context without the trace. However, D1 reveals competing mechanisms rather than a simple confound: the answer distribution and the trace context exert opposing effects on discrimination (§3.7).

1.3 The format-dependence lesson

Cacioli (2026c) showed that switching quantisation format restructures M-ratio profiles while $AUROC_2$ profiles remain perfectly stable. Thinking vs non-thinking mode is analogous to a format change: the same weights produce outputs under different generative regimes. This motivates elevating $AUROC_2$ to co-primary with M-ratio and interpreting any M-ratio result alongside $AUROC_2$.

2 Method

2.1 Pre-registration

The full protocol was filed on OSF (30 April 2026) after 12 independent adversarial reviews across three rounds. The pre-registration specifies co-primary outcomes, four binding diagnostics, exclusion criteria, and a decision table. Two deviations were documented before data collection (Appendix A).

2.2 Models

Model	Type	Parameters	Role
Qwen3-8B	Native hybrid (RL-trained)	8B	Primary
DeepSeek-R1-Distill-Qwen-7B	Distilled from R1-671B	7B	Robustness
DeepSeek-R1-Distill-Llama-8B	Distilled from R1-671B	8B	Robustness

All models: Q5_K_M quantisation, llama-cpp-python (Vulkan backend), AMD RX 7900 GRE (16 GB VRAM). A Q8 quantisation check on 200 items for Qwen3-8B was pre-committed.

2.3 Task and data

2,000 items from TriviaQA rc.nocontext validation split, sampled deterministically (seed 42). Domain distribution: History & Politics 50.2%, Arts & Literature 25.2%, Geography 11.8%, Sports & Games 6.5%, Science & Technology 6.2%. Each model answered all items in both thinking and non-thinking mode.

2.4 Generation parameters (locked)

Temperature 0.6, top_k 20, top_p 0.95, max thinking tokens 4,096, max answer tokens 256, seed = item_index.

2.5 Measures

Per-trial: answer text, correctness (exact match + Levenshtein >= 0.85 + containment fallback; Appendix B), answer NLP (mean token log-probability of answer tokens), first-token softmax probability, trace length (thinking mode), abridged NLP (D1).

Per-condition aggregates: accuracy, $AUROC_2$ (non-parametric, unbinned), d' (K=4 equal-quantile bins from non-thinking NLP, fixed across conditions), meta-d' (metadpy 0.1.2 MLE, Hautus correction), M-ratio (meta-d'/d'), ECE (10-bin, equal-width on min-max rescaled NLP).

2.6 Analysis plan

Confirmatory. H1a: Delta-$AUROC_2$!= 0 (paired bootstrap, 10,000 resamples, seed 42). H1b: Delta-M-ratio != 0 (same). H3: Delta-ECE replicating Yoon et al. (2025).

Binding diagnostics. D1: abridged-NLP $AUROC_2$ and M-ratio. D2: accuracy-based d' stress test. D3: NLP monotonicity (Spearman rho > 0.10 and monotonic quartile accuracy). D4: validity screening protocol (Cacioli, 2026e; arXiv:2604.17714).

Interpretation. H1a/H1b results are only interpreted if D3 and D4 pass in both modes. H1b additionally requires D1.

3 Results

3.1 Accuracy

Thinking mode improved accuracy for Qwen3-8B (55.1% -> 65.7%, +10.6 pp) and marginally for R1-Distill-Llama-8B (52.6% -> 54.4%, +1.8 pp). R1-Distill-Qwen-7B showed no improvement (26.8% -> 24.9%, -1.9 pp).

3.2 Binding diagnostics

D3 (NLP monotonicity). Qwen3-8B passed in both modes (rho = 0.562 non-thinking, 0.155 thinking). R1-Distill-Qwen-7B failed in thinking mode (rho = -0.071, non-monotonic). R1-Distill-Llama-8B failed in thinking mode (rho = 0.068, below 0.10 threshold). Per the pre-registered decision rule, only Qwen3-8B contributes confirmatory evidence for H1a and H1b. The D3 failures are not merely exclusion criteria -- they are themselves evidence that thinking mode can push models across a validity boundary, from discriminative to non-monotonic or inverted confidence (§4.3).

D4 (Validity screening). All non-thinking conditions classified Valid. R1-Distill-Qwen-7B thinking mode classified Invalid (Fp = 0.535, RBS = +0.047). All other thinking conditions classified Valid.

3.3 H1a: $AUROC_2$ (confirmatory)

NLP-based $AUROC_2$ significantly decreased in thinking mode for all three models. The Qwen3-8B result is confirmatory (D3 passes); the R1-Distill results are descriptive (D3 fails).

Model	$AUROC_2$ NT	$AUROC_2$ T	Delta-$AUROC_2$	95% CI	Status
Qwen3-8B	0.826	0.594	-0.231	[-0.262, -0.198]	Confirmatory*
R1-Distill-Qwen-7B	0.670	0.453	-0.218	[-0.257, -0.179]	D3 excluded
R1-Distill-Llama-8B	0.655	0.539	-0.116	[-0.153, -0.088]	D3 excluded

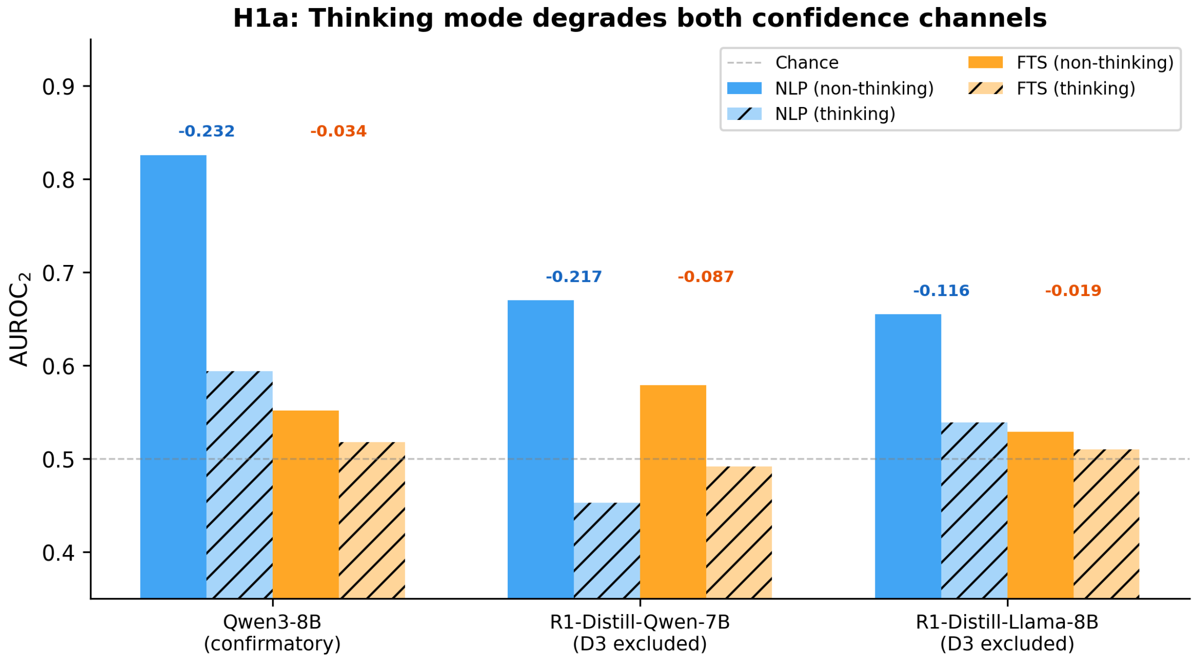

Figure 1. $AUROC_2$ for NLP and first-token softmax (FTS) across models and modes. Both confidence channels degrade in thinking mode. Hatched bars = thinking mode. Delta values shown above each pair.

3.4 First-token softmax $AUROC_2$

First-token softmax probability -- a second token-probability-derived confidence signal -- also degraded in all three models:

Model	Delta-$AUROC_2$(FTS)	Delta-$AUROC_2$(NLP)	Both degrade
Qwen3-8B	-0.034	-0.231	Yes
R1-Distill-Qwen-7B	-0.087	-0.217	Yes
R1-Distill-Llama-8B	-0.019	-0.116	Yes

The first-token effect is substantially smaller than the NLP effect. The convergence across two token-probability-derived signals reduces the likelihood that the effect is specific to mean log-probability, though both remain dependent on the same underlying generative distribution. The larger degradation in mean NLP relative to first-token softmax suggests that compression operates primarily over the token sequence rather than at the initial decision point, consistent with a trajectory-level smoothing effect induced by the thinking trace.

3.5 H1b: M-ratio

M-ratio was not computable for Qwen3-8B and R1-Distill-Qwen-7B in thinking mode (metadpy MLE did not converge; d' = NaN). The cause is distributional: when non-thinking bin boundaries are applied to thinking-mode NLP, 74-79% of trials fall in the highest bin, leaving insufficient distributional structure for the Type-2 SDT model. Thinking mode induces a low-variance regime in which the assumptions required for meta-d' estimation -- particularly stable confidence stratification -- are violated. This extends the format-dependence finding from Cacioli (2026c): $AUROC_2$ remains computable under distributional regime shifts; M-ratio does not, revealing a fragility of parametric Type-2 SDT under distribution shift. R1-Distill-Llama-8B showed M-ratio = 0.985 (thinking) vs 1.257 (non-thinking), Delta-M-ratio = -0.272, but the bootstrap CI included zero [-1.773, +1.099].

3.6 H3: ECE replication

Qwen3-8B showed lower ECE in thinking mode (0.321 vs 0.350, Delta-ECE = -0.025, CI [-0.046, -0.005]), partially replicating Yoon et al. (2025). R1-Distill-Llama-8B showed *higher* ECE in thinking mode (+0.063, CI [+0.026, +0.105]). R1-Distill-Qwen-7B showed no change. The Qwen3-8B result confirms the calibration-discrimination dissociation: improved calibration coexists with degraded discrimination.

3.7 The compression mechanism

The $AUROC_2$ degradation is explained by compression of the NLP distribution in thinking mode.

Model	Var(NLP) NT	Var(NLP) T	Ratio	Cohen's *d* NT	Cohen's *d* T
Qwen3-8B	0.0220	0.0036	0.16	1.023	0.142
R1-Distill-Qwen-7B	0.0758	0.0122	0.16	0.516	-0.124
R1-Distill-Llama-8B	0.0218	0.0131	0.60	0.531	0.104

For Qwen3-8B, NLP variance drops to 16% of non-thinking levels. Cohen's *d* for correct/incorrect separation collapses from 1.02 to 0.14. Both correct and incorrect NLP distributions shift upward, but the reduction in variance -- not ceiling saturation alone -- drives the collapse in separability. The thinking trace produces answers with uniformly high token probabilities regardless of correctness. For R1-Distill-Qwen-7B, the separation actually inverts (*d* = -0.12), consistent with its D4 Invalid classification.

Length-controlled analysis. NLP was residualised on answer length via linear regression and $AUROC_2$ recomputed. The effect strengthened for Qwen3-8B (Delta-$AUROC_2$ = -0.302 length-controlled vs -0.232 raw) and was unchanged for R1-Distill-Llama-8B (-0.117 vs -0.116). Answer length does not account for the compression.

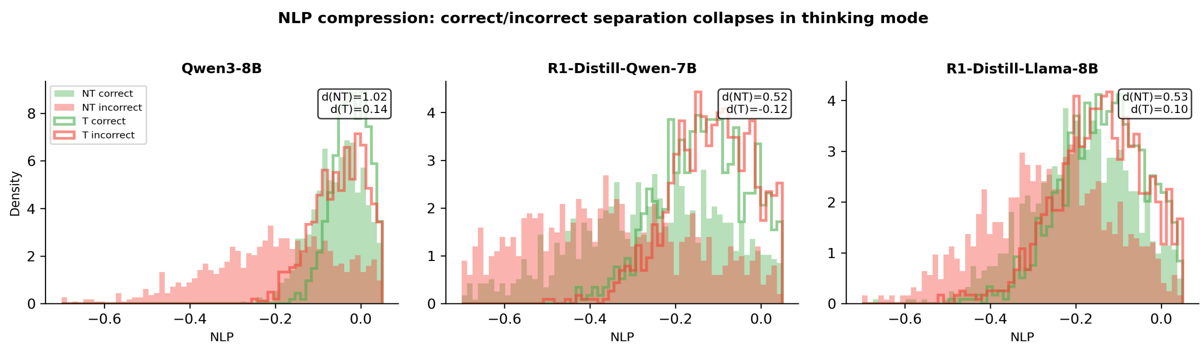

Figure 2. NLP distributions for correct (green) and incorrect (red) trials. Filled histograms = non-thinking; outlined = thinking. Cohen's *d* annotated. Thinking mode compresses both distributions upward, collapsing separability.

3.8 D1: Competing mechanisms

The D1 abridged-NLP diagnostic reveals that thinking mode introduces two opposing forces on NLP discrimination, with model-dependent balance:

Model	Baseline (NT)	Abridged (T ans, NT ctx)	Thinking (T ans, T ctx)	Answer-dist effect	Context effect
Qwen3-8B	0.826	0.455	0.594	-0.371 (160%)	+0.139 (-60%)
R1-Distill-Qwen-7B	0.670	0.659	0.452	-0.011 (5%)	-0.207 (95%)
R1-Distill-Llama-8B	0.655	0.627	0.539	-0.028 (24%)	-0.088 (76%)

Percentages indicate each component's contribution to the total Delta-AUROC$_2$. This decomposition is descriptive rather than strictly causal, as answer generation and context conditioning are not independent processes; however, the pattern cleanly distinguishes dominant pathways across models.[1]

[1] Percentages can exceed 100% when components act in opposing directions; values indicate relative contribution to the net change.

Qwen3-8B shows a dominant answer-distribution effect: the thinking-mode *answers themselves* carry dramatically less discriminative NLP than non-thinking answers, even when evaluated in fresh context without the trace. The trace context then *partially recovers* discrimination (+0.139), but not enough to offset the answer-distribution degradation. The net effect is -0.231.

R1-Distill models show the opposite pattern. The answers are minimally different in discriminability (-0.011 and -0.028). Instead, the trace context itself *actively degrades* discrimination (-0.207 and -0.088). The thinking trace conditions the model into a regime where NLP loses its relationship with correctness.

These are mechanistically distinct failure modes producing the same net result: degraded NLP discrimination under thinking mode.

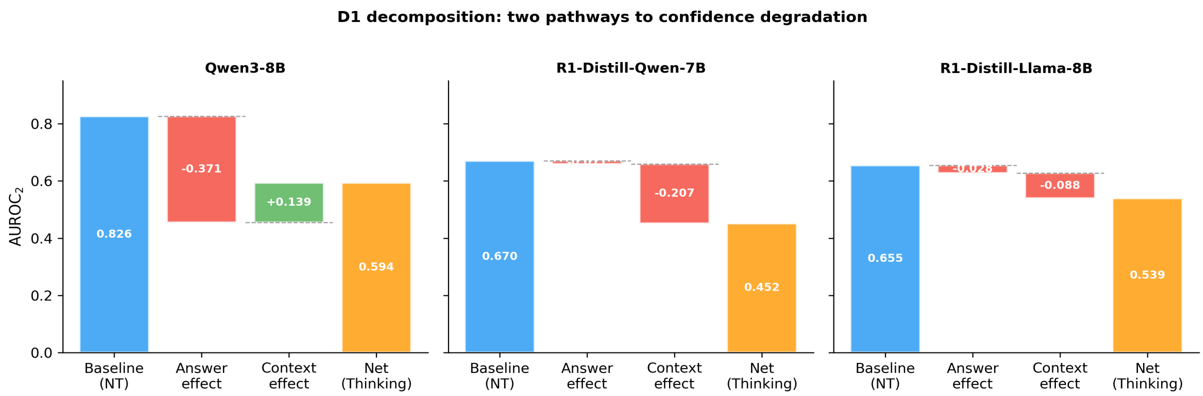

Figure 3. D1 decomposition: waterfall from baseline (non-thinking) AUROC$_2$ through answer-distribution effect and context recovery to net thinking AUROC$_2$. Qwen3-8B: answer-distribution dominant. R1-Distill: context-conditioning dominant.

3.9 Robustness checks

Scoring sensitivity. Under strict exact-match scoring (no containment fallback), the Qwen3-8B Delta-AUROC$_2$ remained negative (-0.122 strict vs -0.232 containment). For R1-Distill models, strict scoring produced insufficient correct trials in non-thinking mode (0.4-1.1% accuracy) for meaningful AUROC$_2$ computation, confirming the containment fallback is necessary for these models.

K-robustness. Delta-AUROC$_2$ was invariant to bin count (K=3, 4, 6), as expected for an unbinned metric.

Q8 quantisation. On 200 matched items, Q5_K_M AUROC$_2$ = 0.825, Q8 AUROC$_2$ = 0.798. NLP Spearman correlation = 0.931 (p < 10■■■). Sign consistent; magnitude within +/-50%.

Cross-model consistency (E6). All three models showed negative Delta-AUROC$_2$, consistent across native (Qwen3) and distilled (R1) reasoning architectures.

3.10 Exploratory results

E1 (Difficulty interaction). The AUROC$_2$ degradation was strongest for hard items (Qwen3-8B: Delta- = -0.172) and absent for easy items (Delta- = +0.002). Thinking mode reduces confidence discrimination most where it is needed most.

E2 (Trace length-accuracy). Strongly negative across all models (Qwen3-8B: rho = -0.414, p < 10■■■). Longer traces predict incorrect answers.

E3 (Trace length-confidence). Overall rho was negative for Qwen3-8B (-0.203) and R1-Distill-Llama-8B (-0.132, replicating the saturation paper's directional prediction for R1-Distill models). R1-Distill-Qwen-7B showed a weak positive correlation (+0.087), not replicating.

E4 (Trace asymmetry). Error traces were 1.2-2.1x longer than correct traces. The ratio increased with item easiness.

E5 (Structural features). Backtracking markers appeared in 77-92% of traces. Items with backtracking were slightly less accurate than baseline (-2 to -3 pp), suggesting backtracking reflects genuine uncertainty rather than successful self-correction.

H4 (Domain interaction). Domain-level AUROC$_2$ variance did not exceed chance (permutation p = 0.07-0.33).

4 Discussion

4.1 Summary

Thinking mode significantly reduces the discriminative power of token-probability-derived confidence signals in reasoning LLMs. The effect is large (Delta-AUROC$_2$ = -0.231 for the primary model), consistent across two token-level signals (NLP and first-token softmax), robust to confounds (answer length, quantisation, scoring method), and replicated across model families. Across models and diagnostics, thinking mode induces a confidence homogenisation regime: responses become more accurate on average but less separable in confidence space.

Three hypotheses can account for the observed degradation: (A) a *channel artefact* -- the generative regime changes the statistical properties of NLP without changing the model's internal uncertainty representations; (B) an *answer-distribution shift* -- thinking-mode answers are intrinsically less discriminable by token-level signals; (C) *internal metacognitive degradation* -- the model's uncertainty representation itself is compressed. Our evidence constrains but does not fully resolve this space. The D1 decomposition supports (B) for Qwen3-8B (answer-distribution dominant) and a mixture of (A) and (B) for R1-Distill models (context-conditioning dominant). The first-token softmax convergence makes a pure (A) interpretation less likely, since two token-level signals degrade in parallel. Hypothesis (C) would require internal-state probing, which we leave to future work.

4.2 Compression and competing mechanisms

For Qwen3-8B, the dominant mechanism is answer-distribution shift: thinking-mode answers have inherently less discriminative NLP, even in fresh context. The model's reasoning process produces answers that are uniformly well-predicted token-by-token regardless of correctness. The trace context partially recovers discrimination, suggesting the trace carries some residual uncertainty information -- but not enough to offset the answer-level compression.

For R1-Distill models, the mechanism is different: the answers retain their discriminability, but the trace context actively degrades it. The extended thinking sequence creates a local context that makes even incorrect answers fluent and well-predicted.

Both pathways produce the same observable outcome -- compressed NLP, degraded $AUROC_2$ -- but through distinct generative processes. This distinction matters for mitigation: for Qwen3-8B-like models, the fix requires changing how thinking mode produces answers; for R1-Distill-like models, the fix requires extracting confidence without trace conditioning.

4.3 D3 failures as findings

The D3 monotonicity failures in R1-Distill models are not merely exclusion criteria -- they indicate that thinking mode induces regimes where NLP no longer satisfies minimal validity criteria as a confidence signal. R1-Distill-Qwen-7B shows inverted confidence (rho = -0.071): higher NLP is associated with *lower* accuracy. This is consistent with the D4 Invalid classification for that condition and with prior findings on reasoning-mode validity failure in the CMM programme (Cacioli, 2026e). Similarly, M-ratio non-convergence under thinking mode reflects fragility of parametric Type-2 SDT under distributional regime shift: the assumptions of stable confidence stratification are violated when NLP variance collapses. $AUROC_2$ remains computable; M-ratio does not -- extending the format-robustness finding from Cacioli (2026c).

4.4 Calibration versus discrimination

Yoon et al. (2025) showed that reasoning models produce better-calibrated *verbalized* confidence -- the confidence score the model states in words. We partially replicate this for ECE on Qwen3-8B (Delta-ECE = -0.025). However, we find the opposite pattern for *implicit* confidence: the token-probability signals that the model does not deliberately produce become less discriminative under thinking mode. This dissociation between verbal and implicit channels is striking. It is consistent with Miao and Ungar's (2026) finding that accuracy and verbalized confidence occupy orthogonal directions in the residual stream -- the model can learn to verbalise better confidence through slow thinking (Yoon et al., 2025) while simultaneously compressing the implicit token-probability signal that it does not explicitly control. For applications requiring per-trial reliability assessment (selective prediction, deferral), the implicit signal is often the only channel available, making its degradation practically consequential.

4.5 Relationship to prior work

Verbal saturation. Cacioli (2026d) found verbal confidence saturated at ceiling in 3-9B models -- a readout failure at the output interface. The present finding extends this to implicit signals: NLP compresses under thinking mode, not because of a readout failure, but because the generative regime itself changes the distributional geometry of the confidence signal. The saturation paper documented interface failure; this study documents generative-regime failure.

Reasoning contamination. Miao and Ungar (2026) showed that when models solve a problem and rate their confidence jointly, the confidence and accuracy directions in activation space shift from weakly aligned to sharply opposed (cosine similarity dropping from +0.26 to -0.63) -- a phenomenon they termed the Reasoning Contamination Effect. Our NLP compression finding is consistent with a related but distinct mechanism operating at the token-probability level: the thinking trace compresses the implicit confidence distribution rather than inverting it. The negative trace length-confidence correlation (E3) replicates the saturation paper's directional prediction for one of two R1-Distill variants. Longer reasoning compresses NLP further, compounding the cost.

4.6 Practical implications

For any application that uses implicit confidence signals -- selective prediction, abstention, deferral to human experts -- switching from non-thinking to thinking mode degrades signal quality. Practitioners face a trade-off: thinking mode improves accuracy but makes it harder to identify *which* answers are correct. The D1 result further suggests that the obvious mitigation -- recomputing confidence in fresh context -- does not fully recover discrimination, because the thinking-mode answers themselves carry less discriminative signal.

4.7 Threats to validity

Signal vs metacognition. We have shown that token-probability-derived confidence signals degrade, not that the model's internal metacognitive state is degraded. Probes trained on internal activations during the thinking trace may reveal preserved or even enhanced uncertainty representations that the token-level confidence signals fail to transmit. The convergence across two token-level signals reduces the likelihood that the effect is specific to mean log-probability, though both remain dependent on the same underlying generative distribution and could degrade for the same non-metacognitive reason.

Generative regime invariance. NLP's statistical meaning changes between regimes. The thinking trace fundamentally alters the conditioning context, entropy structure, and token predictability. Our $AUROC_2$ result establishes that the *ranking* of correct vs incorrect responses by NLP degrades, which is operationally meaningful for applications. Whether this reflects a genuine metacognitive change or a measurement artefact of the regime shift is a question our design can constrain but not resolve.

Scale. Three 7-8B models on one benchmark. Stage 2 (27-32B models on M3 Ultra hardware) will test whether the compression persists at larger scale, where thinking traces may carry more structured information and models may possess circuit redundancy for maintaining secondary confidence signals.

Benchmark. TriviaQA is predominantly factual recall. This study tests reasoning-mode *mechanics* rather than reasoning-task *benefits*. On factual recall, thinking traces often function as retrieval rehearsal rather than genuine multi-step inference. Reasoning-heavy benchmarks (mathematics, formal logic) may show different patterns if thinking traces engage qualitatively different processes -- the degradation documented here may not generalise to tasks where the trace performs genuine computation.

Consumer hardware. All inference ran on an AMD RX 7900 GRE (Vulkan backend). The logprobs=1 parameter, necessitated by a performance bug in llama-cpp-python's full-vocabulary logprob sorting, returns top-1 log-probabilities only. Entropy-based confidence channels, which require the full vocabulary distribution, could not be evaluated. This is an important limitation given that entropy may be more robust to distributional compression than mean log-probability.

5 Conclusion

Thinking mode in reasoning LLMs improves accuracy but reduces the discriminative power of token-probability-derived confidence signals. The evidence is consistent with distributional compression as the primary mechanism, operating through model-dependent pathways:

answer-distribution shift in natively trained reasoning models, and context-conditioning degradation in distilled models. Two token-level confidence signals (NLP and first-token softmax) degrade in parallel, with compression operating primarily at the sequence level rather than the initial decision point. Parametric Type-2 SDT (M-ratio) becomes ill-posed under the induced distributional regime, while non-parametric $AUROC_2$ remains computable -- extending the format-robustness finding from prior work.

The practical implication is a trade-off between accuracy and reliability monitoring. For practitioners building selective-prediction or deferral systems, we suggest: (a) use thinking mode for answer generation; (b) compute confidence via a separate non-thinking pass or a learned aggregator combining multiple signals; (c) validate confidence discriminability on held-out data before deployment. Better answers and worse confidence discrimination are not contradictory -- they are two consequences of the same generative process.

References

Cacioli, J.-P. (2026a). Beyond the mean: Type-2 signal detection theory for LLM metacognition. *arXiv:2603.25112*.

Cacioli, J.-P. (2026b). Signal detection theory for large language models. *arXiv:2603.14893*.

Cacioli, J.-P. (2026c). Quantisation and metacognition in LLMs. *arXiv:2604.08976*.

Cacioli, J.-P. (2026d). Verbal confidence saturation in 3-9B open-weight instruction-tuned LLMs. *arXiv:2604.22215*.

Cacioli, J.-P. (2026e). A validity screening protocol for LLM confidence signals. *arXiv:2604.17714*.

DeepSeek-AI. (2025). DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv:2501.12948*.

Fleming, S. M., & Lau, H. C. (2014). How to measure metacognition. *Frontiers in Human Neuroscience, 8*, 443.

Maniscalco, B., & Lau, H. (2012). A signal detection theoretic approach for estimating metacognitive sensitivity from confidence ratings. *Consciousness and Cognition, 21*(1), 422-430.

Miao, M. M., & Ungar, L. (2026). Closing the confidence-faithfulness gap in large language models. *arXiv:2603.25052*.

Qwen Team. (2025). Qwen3 technical report. https://qwenlm.github.io/blog/qwen3/

Yoon, D., Kim, S., Yang, S., Kim, S., Kim, S., Kim, Y., Choi, E., Kim, Y., & Seo, M. (2025). Reasoning models better express their confidence. *NeurIPS 2025*. arXiv:2505.14489.

Appendix A: Deviation log

Deviation 1 (1 May 2026). R1-Distill models produced empty thinking traces on TriviaQA items (0-4% trace generation rate). Added `\n` assistant prefill per DeepSeek's official recommendation. Classified as permissible non-deviation under §10.1.

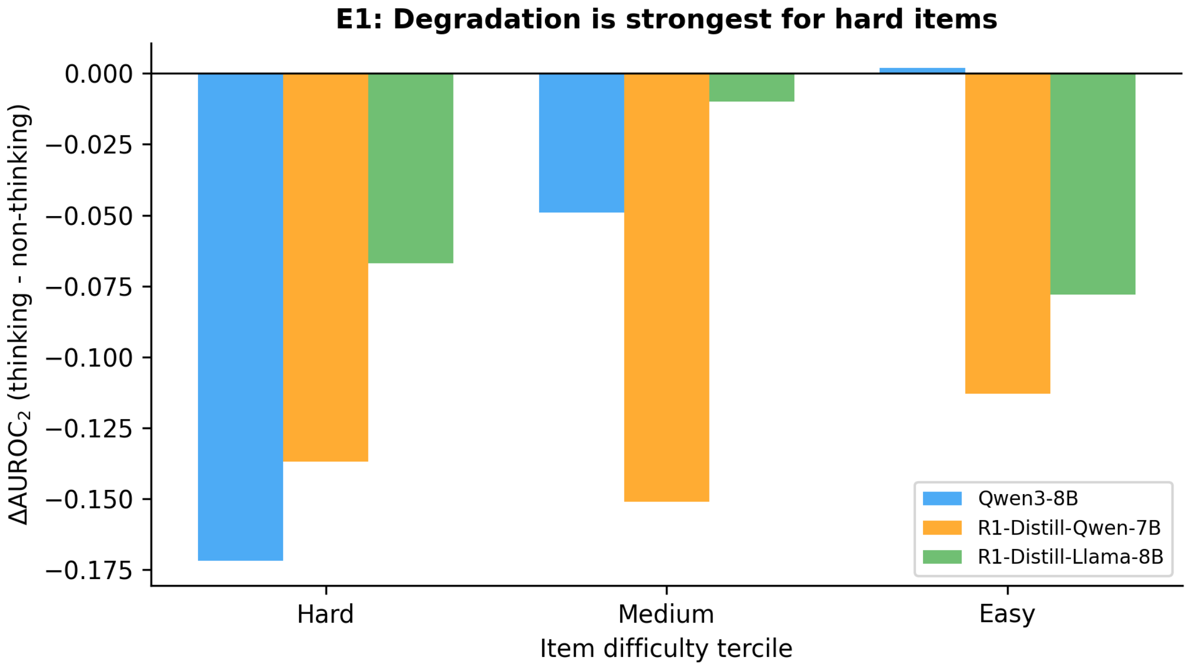

Figure 4. E1: Delta-AUROC$_2$ by item difficulty tercile. Degradation is strongest for hard items and absent for easy items in Qwen3-8B.

Deviation 2 (2 May 2026). R1-Distill-Qwen-7B produced 1.2% accuracy in non-thinking mode due to chain-of-thought contamination as plain text. Added `\n\n` empty prefill and containment scoring fallback. Both documented before full data collection.

Appendix B: Containment scoring

Exact match + Levenshtein >= 0.85 produced systematically low accuracy for R1-Distill models that generate verbose responses embedding correct answers in longer text. A containment fallback was added: after exact match and Levenshtein fail, the scorer checks whether any normalised gold alias (>= 4 characters, parenthetical disambiguators stripped) appears as a substring in the normalised response. This follows standard TriviaQA evaluation practice and was applied uniformly across all models and conditions. Under strict exact-match scoring (no containment), the Qwen3-8B Delta-AUROC$_2$ remained negative (-0.122), confirming the primary result is not driven by the scoring method.

Appendix C: Scoring sensitivity

Model	Delta-AUROC$_2$ (containment)	Delta-AUROC$_2$ (strict)	Finding stable
Qwen3-8B	-0.232	-0.122	Yes
R1-Distill-Qwen-7B	-0.217	+0.230	No (insufficient strict accuracy)
R1-Distill-Llama-8B	-0.116	+0.418	No (insufficient strict accuracy)

The sign reversal for R1-Distill models under strict scoring reflects floor effects: strict accuracy is 0.4-1.1% in non-thinking mode, too low for meaningful AUROC$_2$. The containment fallback is necessary for these models to have sufficient correct trials for analysis. The Qwen3-8B result -- the only confirmatory model -- is stable under both scoring methods.